

Domain and Range from Problem Situations

Algebra II | Unit 1 | Day 2 | TEKS A2.7.I, A2.2.A

Objective: I can find the domain and range of a real-world situation, explain what they mean in context, and tell the difference between the domain of a function and the domain that the situation allows.

Vocabulary

Independent variable — the _____, the one you choose. Goes on the _____-axis.

Dependent variable — the _____, it depends on the input. Goes on the _____-axis.

Function domain — every input the _____ will accept.

Contextual domain — every input the _____ actually allows. Usually _____.

Part 1 — Name the Variables

Which quantity is the input and which is the output?

1. Hours you work and the money you earn

independent: _____ dependent: _____

2. Number of tickets bought and the total cost

independent: _____ dependent: _____

3. Seconds since a ball was thrown and its height

independent: _____ dependent: _____

Part 2 — Function Domain vs. Contextual Domain

A cube-shaped box has side length x inches. Its volume is $V(x) = x^3$.

What domain does the **equation** x^3 accept? _____

What domain does the **box** allow? _____

Why? _____

Big idea: the equation may accept numbers the _____ does not. Always ask what makes sense.

Part 3 — Domain and Range in Context

Give the contextual domain and range. Use an inequality and interval notation.

Example 1. Concert tickets cost \$12 each. You may buy at most 8. Total cost $C(t) = 12t$.

Domain: _____ Discrete or continuous? _____

Range: _____

You Try 1. A ball is thrown. Its height is $h(t) = -16t^2 + 64t$ feet, and it is in the air from $t = 0$ to $t = 4$ seconds. Its greatest height is 64 feet.

Domain: _____ interval _____ Discrete or continuous? _____

Range: _____ interval _____

You Try 2. A parking garage charges \$3 per hour and is open 6 a.m. to 10 p.m. (16 hours).

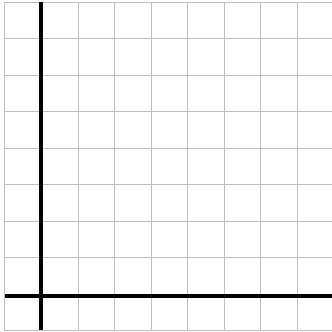
Domain: _____ Range: _____

Part 4 — Reading a Situation from a Graph

Each square is 1 unit. Plot the points, then answer in context.

A water tank is drained. Gallons g after m minutes:

m	0	1	2	3	4
g	8	6	4	2	0



Answer in context.

Domain: _____

Range: _____

What does the domain mean here?

What does a range of 0 mean?

Part 5 — Does the Situation Restrict the Domain?

Write yes or no, and if yes, give the restriction.

- The side length of a square garden _____
- The temperature in a freezer over one day _____
- The number of buses needed for a field trip _____

Practice

Give the contextual domain and range for each. Show your thinking on the lines.

- A phone plan costs \$30 per month plus \$5 per gigabyte, up to 20 GB.

Domain _____ Range _____

- A theater has 250 seats. Tickets are \$9 each, and t tickets are sold.

Domain _____ Range _____

- A diver descends from the surface to 40 meters below sea level.

Range of depth _____ Why negative? _____

Exit Ticket

A taxi charges a \$4 flat fee plus \$2 per mile, for trips up to 10 miles. Cost $C(m) = 4 + 2m$.

Domain _____ Range _____

Why does the range start at 4, not 0? _____